

SECONDBRAIN - CLAUDE CO-WORK HANDOVER

PROJECT

Name: SecondBrain

Repository: hf-personal-digital-vault

Core Question:

Can Hugging Face Buckets become my personal digital memory system?

AUTHOR CONTEXT

- 20+ years enterprise infrastructure experience
- Cloud and virtualization background
- Learning AI/ML during career break
- Learns by building systems from first principles

PROJECT PHILOSOPHY

1. Chat history is ephemeral. Documentation is persistent.
2. Architecture first.
3. Keep production code and experiments separate.

CURRENT STATUS

Day: 02

Phase: Phase 1B – SQLite Catalog Design

COMPLETED

Day 01:

- Installed uv
- Initialized git/project
- Authenticated HF
- Created HF Bucket
- Uploaded first object

Bucket:

hinabarun/testproject

Object:

secondbrain-folder/hello-secondbrain.txt

Day 02:

Asset lifecycle:

DISCOVERED -> HASHING -> DEDUP_CHECK -> PROCESSING -> UPLOADING -> VERIFYING -> VERIFIED

Terminal states:

DUPLICATE, FAILED

Duplicate strategy:

A file is duplicate only if:

- SHA256 matches
- existing asset status == VERIFIED

Asset identity:

AssetId = str (ULID planned)

Asset schema:

id
asset_type
storage_status
failure_stage
duplicate_of
filename
local_path
title
description
tags
bucket_name
object_key
mime_type
size_bytes
sha256
file_modified_at
original_date
metadata
thumbnail_path
embedding_id

MOST IMPORTANT DECISION

SQLite = Source of Truth

HF Buckets = Blob Storage Backend

SQLite owns:

- metadata
- upload state
- deduplication
- provenance
- workflow tracking

HF Buckets own:

- blobs
- thumbnails
- future artifacts

ROADMAP

Current Phase 1B tasks:

- Design SQLite schema
- Design indexes
- Build repository layer
- Create SQLite database
- Persist first asset
- Retrieve first asset

YOUR ROLE

Act as:

- senior software architect
- senior Python engineer
- mentor
- technical co-author

Before writing code:

- critique architecture
- identify missing decisions
- propose alternatives
- explain tradeoffs
- then implement.

Treat this as a production-grade system and future technical blog series.